

AEGNN-M: A 3D Graph-Spatial Co-Representation Model for Molecular Property Prediction

Lijun Cai , Yuling He , Xiangzheng Fu , Linlin Zhuo , Quan Zou , Senior Member, IEEE, and Xiaojun Yao 

Abstract—Improving the drug development process can expedite the introduction of more novel drugs that cater to the demands of precision medicine. Accurately predicting molecular properties remains a fundamental challenge in drug discovery and development. Currently, a plethora of computer-aided drug discovery (CADD) methods have been widely employed in the field of molecular prediction. However, most of these methods primarily analyze molecules using low-dimensional representations such as SMILES notations, molecular fingerprints, and molecular graph-based descriptors. Only a few approaches have focused on incorporating and utilizing high-dimensional spatial structural representations of molecules. In light of the advancements in artificial intelligence, we introduce a 3D graph-spatial co-representation model called AEGNN-M, which combines two graph neural networks, GAT and EGNN. AEGNN-M enables learning of information from both molecular graphs representations and 3D spatial structural representations to predict molecular properties accurately. We conducted experiments on seven public datasets, three regression datasets and 14 breast cancer cell line phenotype screening datasets, comparing the performance of AEGNN-M with state-of-the-art deep learning methods. Extensive experimental results demonstrate the satisfactory performance of the AEGNN-M model. Furthermore, we analyzed the performance impact of different modules within AEGNN-M and the influence of spatial structural representations on the

model's performance. The interpretability analysis also revealed the significance of specific atoms in determining particular molecular properties.

Index Terms—Drug development, precision medicine, molecular properties, deep learning, molecular representation, drug discovery.

I. INTRODUCTION

PRECISION medicine holds the potential to enhance the accuracy and efficacy of disease treatment [1]. Drug development plays a pivotal role in the field of precision medicine [2]. Molecular property prediction plays a role in drug discovery and development by accurately forecasting the characteristics and interactions of molecules [3], thereby expediting the drug screening and optimization processes and providing substantial support for drug development to drive the advancement of precision medicine [4]. Drug development is a lengthy, resource-intensive, and high-risk process. Many traditional Computer-Aided Drug Design (CADD) methods based on molecular simulation techniques incur significant computational costs and time-consuming processes.

In recent years, several artificial intelligence techniques have been successfully applied to molecular property prediction tasks [5]. Numerous efforts have been made from various directions to design SMILES-based models for predicting molecular properties, such as ConvS2S [6], SCFPs [7], Mol-BERT [8], S2DV [9]. However, using only a one-dimensional representation yields very limited molecular information that can be learned. Graph-based models for molecular representation began to rise in popularity, such as DMPNN [10], SGCN [11], MAT [12], CoMPT [13]. Although numerous computational methods have been proposed for predicting molecular properties, the majority of current approaches only utilize low-dimensional representations of molecules, the critical geometric information of molecules remains rarely explored. These low-dimensional representations may limit the amount of information available for computational methods. Therefore, it is imperative to develop effective approaches that integrate molecular spatial structure representations to predict molecular properties, an area of research that has been few investigated before.

To alleviate the aforementioned limitations, we propose a 3D graph-spatial co-representation model, AEGNN-M, for

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Lijun Cai and Yuling He are with the College of Computer Science and Electronic Engineering, Hunan University, Changsha 410012, China (e-mail: ljcai@hnu.edu.cn; heyuling@hnu.edu.cn).

Xiangzheng Fu is with the Research Institute of Hunan University in Chongqing, Chongqing 401120, China (e-mail: fxz326@hnu.edu.cn).

Linlin Zhuo is with the College of Data Science and Artificial Intelligence, Wenzhou University of Technology, Wenzhou 325000, China (e-mail: zhuoninnin@hnu.edu.cn).

Quan Zou is with the Institute of Fundamental and Frontier Sciences, University of Electronic Science and Technology of China, Chengdu 610054, China (e-mail: zouquan@nclab.net).

Xiaojun Yao is with the Centre for Artificial Intelligence Driven Drug Discovery, Faculty of Applied Sciences, Macao Polytechnic University, Macao 999078, China (e-mail: xjyao@mpu.edu.mo).

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molecular property prediction, taking a step towards incorporating both molecular graph representation and spatial information. The AEGNN-M framework integrates two graph neural networks, Graph Attention Network [14] and Equivariant Graph Neural Network [15]. Specifically, the SMILES [16] string is transformed into molecular graph representation and molecular spatial coordinate representation using the Rdkit [17]. Initially, we employ a graph self-attention network to aggregate the features of the molecular graph representation, which are then combined with the molecular spatial coordinate representation. This fused representation is subsequently fed into an equivariant graph neural network for training. As a result, the updated molecular graph representation and molecular spatial coordinate representation encapsulate richer molecular structural information. Finally, we concatenate the molecular graph representation and spatial coordinate representation and employ a simple neural network to predict downstream tasks.

In summary, the main contributions of this paper are described as follows:

- We propose a 3D graph-spatial co-representation model called AEGNN-M for precise molecular property prediction, which leverages two graph neural networks, the graph self-attention model and the equivariant graph neural network, for training molecular representations.
- We combine the molecular graph representation and 3D molecular spatial coordinate representation to capture a comprehensive range of information for training.
- We conduct extensive experiments using benchmark datasets. The results show that AEGNN-M achieves competitive performance in predicting molecular properties.

The structure of the remaining parts of this paper is as follows. Section II presents the related works. Section III focuses on the methods. Section IV reports the results of our experiments, and Section V presents our conclusion.

II. RELATED WORK

A. Molecule Representation

In the field of molecular studies, an essential aspect to consider is the appropriate representation of a molecule [18]. As shown in Fig. 1, molecular representations can be categorized into 1D, 2D, and 3D representations [19]. The research on both SMILES-based and graph-based molecular prediction models is extensively conducted and studied. Hou et al. proposed an enhanced bidirectional LSTM with channel and spatial attention networks, improved through Bayesian optimization, specifically designed for identifying elemental factors in SMILES representation [20]. Li et al. utilized a hybrid architecture consisting of stacked CNN and RNN layers to extract representations from SMILES representation [21]. Jiang et al. utilized graph-based deep learning models such as GCN, GAT, MPNN, and Attentive FP to extract information from molecular graph representations [22]. Ma et al. proposed the Cross-Dependent Graph Neural Network, which constructs atomic-centric and bond-centric views and introduces a cross-dependent message-passing scheme between these two views. However [23], Cai et al. introduced the FP-GNN model, which learns information

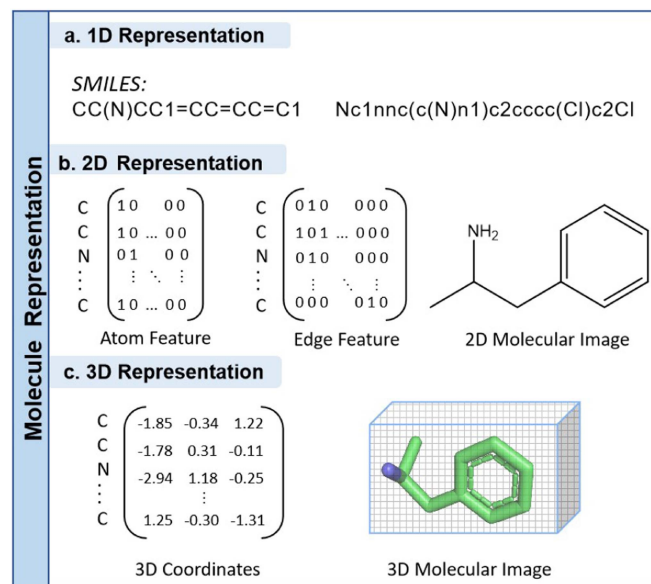


Fig. 1. 1D, 2D, and 3D Molecule Representations.

from both molecular graph representations and molecular fingerprint representations to predict molecular properties [24]. Liu et al. proposed the molecule graph-enhanced model ATMOL+, which leverages molecular graphs representations for molecule representation and property prediction [25]. A significant branch of improvement is effectively utilizing the geometric information in the 3D structure of molecules.

B. Graph Neural Network

Graph neural network (GNN) [26] have become a popular method for predicting molecular properties due to their ability to handle graph-structured data. GCN extends Convolutional Neural Network to handle graph-structured data [27]. MPNN operates on graphs using a message passing scheme [28]. GAT utilizes the attention mechanism to assign varying importance weights to the neighbors of nodes during the aggregation process [14]. EGNN introduces equivariant symmetries of geometric graph spatial features [15]. Graphormer is built directly on top of the standard Transformer architecture and facilitates interactions between distant nodes [29]. These powerful GNN models have been widely applied in various domains of representation learning, including molecular dynamics simulations, protein-small molecule docking, antibody sequence and structural co-design, among others. Inspired by these advancements, we aim to leverage GNN to learn informative representations from molecular structures and achieve strong predictive performance.

III. METHODOLOGY

A. Problem Formulation

Given a molecule set M , prediction task number C . For each molecule $m \in M$, we allocate label p based on the number of tasks that need to be predicted for molecule m . For a single

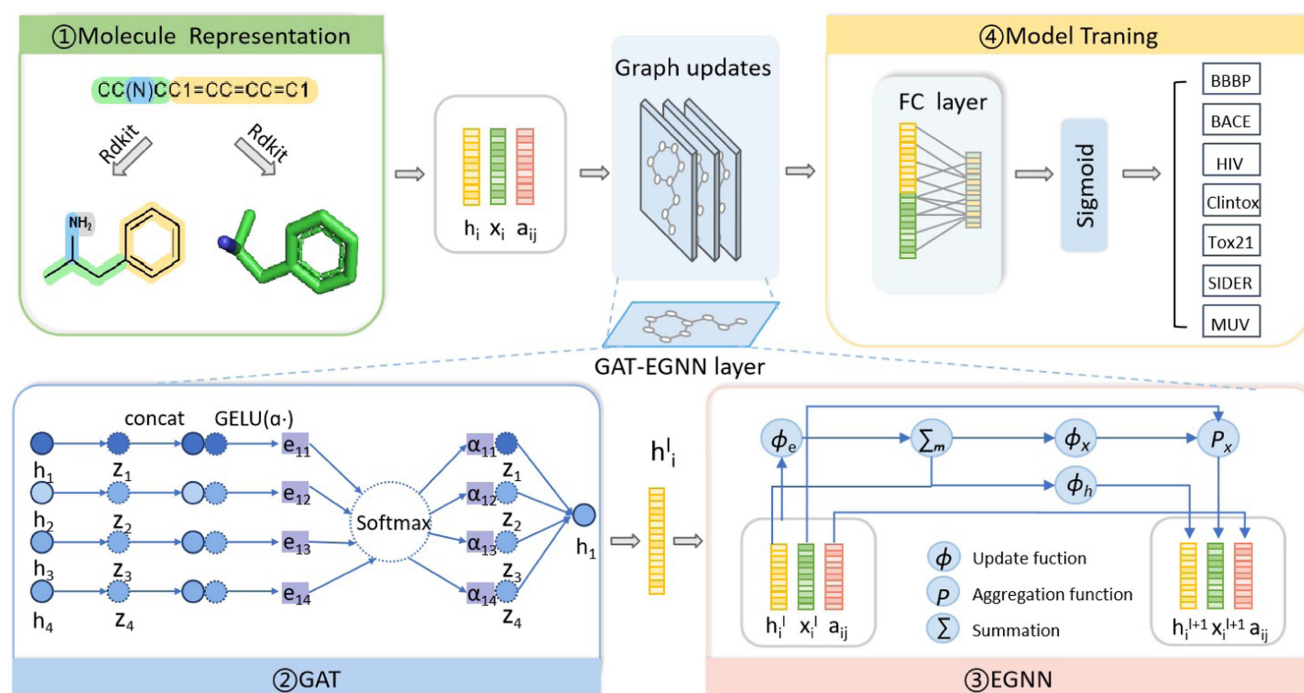


Fig. 2. Workflow of AEGNN-M Network Architecture: ① Initial Molecule Representation, ② Graph Attention Network(GAT), ③ Equivariant Graph Neural Networks(EGNN), ④ Model training.

molecule prediction classification task, there is only one label $p \in \{0, 1\}$. For multiple molecule prediction classification tasks, multiple labels $p_1, p_2, \dots, p_c \in \{0, 1\}$ are assigned, where $p = 1$ indicates that molecule m possesses a certain property, and $p = 0$ indicates the absence of the property. For molecule prediction regression tasks, p represents a specific numerical value label. Our goal is to learn a model that can predict the underlying associations between molecules and the labels representing their properties.

B. Model Architecture

AEGNN-M is an end-to-end deep learning model composed of four parts (Fig. 2): 1) Initial molecule featurization; 2) Graph Attention Network; 3) Equivariant Graph Neural Networks; 4) Model training. In the following parts, we present a detailed description of each stage and provide necessary operational details.

1) AEGNN-M Network Architecture: The AEGNN-M architecture combines 2D molecular graphs representations with molecular 3D coordinate representations to form a flexible network. The simplified molecular input line entry system (SMILES) symbols of the molecules are input into the AEGNN-M architecture. Along the paths, the GAT model captures the information of the 2D molecular graphs. The nodes are aggregated from themselves and their neighbors through self-attention, generating node features h_i and 3D coordinate features x_i , which are then input into the EGNN model. The EGNN model utilizes aggregation functions based on the principle of equivariance for 3D coordinate features. Even when aggregating node features from 2D molecular graph features, the 3D coordinates still

maintain equivariance. The resulting 2D node features h_i^{l+1} and 3D coordinates x_i^{l+1} are averaged and concatenated to represent the molecule. Finally, The concatenated feature are fed into a fully connected layer to generate the final output for matching the downstream task.

2) Initial Molecule Representation: We convert the each SMILES into 2D molecular graph $G(V, E)$ and 3D coordinate that associated with each of the graph nodes by Rdkit. $G(V, A)$ includes nodes $v_i \in V$ which represents atom and edge attributes $a_{ij} \in A$ which represents chemical bonds. For each atomic node i , we perform feature embedding to encode its characteristics into h_i . Similarly, the molecular graph edge a_{ij} , we also apply feature embedding. Additionally, we perform feature embedding on the 3D coordinates of the atoms, represented as x_i .

3) Graph Attention Network(GAT): GAT is an architecture that is based on the multi-head attention mechanism, which updates each node by aggregating information from itself and other nodes. The attention mechanism focus on the most relevant neighbors when aggregating neighbor information. GAT calculates the correlation coefficient between node j and node i according to the following equation:

$$e_{ij} = GELU(a \cdot [Wh_i || Wh_j]). \quad (1)$$

First, the node features are enhanced through a linear mapping using shared parameter W . Subsequently, the transformed features of nodes i and j are concatenated. Next, a learnable vector a is utilized to map the concatenated high-dimensional features to a scalar. Finally, the activation function GELU is applied to obtain the correlation coefficient e_{ij} . The normalization of all

neighbor correlation coefficients can be calculated as follow:

$$\alpha_{ij} = \text{softmax}(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{k \in N(i)} \exp(e_{ik})}. \quad (2)$$

$N(i)$ means the neighbors of node i , and α_{ij} represent the attention coefficient between nodes i and j . The attention coefficient is utilized to update node as follow:

$$h_i = \sigma \left(\alpha_{ii} W h_i + \sum_{j \in N(i)} \alpha_{ij} W h_j \right), \quad (3)$$

where σ is a multi-layer perception that includes the GELU activation function. In each layer of the GAT network, a multi-head attention mechanism is applied with the number of attention heads set to 8.

4) Equivariant Graph Neural Networks(EGNN): EGNN is a model to learn graph neural networks equivariant to rotations, translations, reflections and permutations with respect to an input set of points. EGNN takes as input the edge attributes a_{ij} , 3D coordinate embeddings $x^l = \{x_1^l, x_2^l, \dots, x_n^l\}$, and the node embeddings $h^l = \{h_1^l, h_2^l, \dots, h_n^l\}$ obtained after applying GAT. Our model will preserve the equivalence of rotations and translations on these 3D coordinate set, as well as the equivalence of permutations on the node set. The equations that define EGNN layer are the following:

$$m_{ij} = \phi_e(h_i^l, h_j^l, \|x_i^l - x_j^l\|^2, a_{ij}), \quad (4)$$

$$x_i^{l+1} = x_i^l + C \sum_{j \neq i} (x_i^l - x_j^l) \phi_x(m_{ij}), \quad (5)$$

$$m_i = \sum_{j \in N(i)} m_{ij}, \quad (6)$$

$$h_i^{l+1} = \phi_h(h_i^l, m_i). \quad (7)$$

In (4), we input the node embeddings h_i and h_j of two points, the edge attributes a_{ij} between them, and the relative distance $\|x_i^l - x_j^l\|^2$ between the two points into the edge operation ϕ_e . In (5), the position of each particle, denoted as x_i , is updated using a vector field in a radial direction. This update is achieved by calculating the weighted sum of the relative differences $(x_i - x_j)$ between all particles. The weights for this sum are determined by the function ϕ_x , which takes the edge embedding m_{ij} obtained from the previous edge operation as input and produces a scalar value. To normalize the sum, the value of parameter C is chosen as $1/(M-1)$, where M represents the number of elements in the sum. Despite its simplicity, this operation exhibits great flexibility as the embedding m_{ij} can encapsulate information from the entire graph. Equation (6) aggregates the incoming messages from neighboring nodes represented as $N(i)$ to the node. On the other hand, (7) applies the node operation ϕ_h . This operation takes the aggregated messages m_i , along with the node embedding h_i^l , as input and generates the updated node embedding h_i^{l+1} as output.

5) Model Training: We take the average of the learned embeddings h_i^{l+1} for node features and the embeddings x_i^{l+1} for

TABLE I
CLASSIFICATION TASKS OF THE SEVEN BENCHMARK DATASETS

Dataset	Molecule number	Task number
BBBP	2039	1
BACE	1513	1
HIV	41127	1
Clintox	1478	2
Tox21	7831	12
SIDER	1478	27
MUV	93087	13

3D coordinate representations. The formula is as follows:

$$H = \left(\frac{1}{N} \sum_{i=1}^N h_i^{l+1} \right), \quad (8)$$

$$X = \left(\frac{1}{N} \sum_{i=1}^N x_i^{l+1} \right). \quad (9)$$

For molecular property prediction, we utilize the learned embeddings of 2D molecular graph features H and 3D coordinate representations X . These embeddings are input to a fully connected neural network, followed by a sigmoid function, to output the classification task results $\hat{p}$:

$$\hat{p} = \text{Sigmoid}(FC(H||X)). \quad (10)$$

while the regression task directly yields the results $\hat{p}$:

$$\hat{p} = FC(H||X). \quad (11)$$

For classification task, the loss function during the training process be formulated as:

$$\begin{aligned} Loss = & -\frac{1}{T} \sum_{t \in T} \frac{1}{C} \sum_{c \in C} (p_{t,c} \log(\hat{p}_{t,c}) \\ & + (1 - p_{t,c}) \log(1 - \hat{p}_{t,c})). \end{aligned} \quad (12)$$

For regression task, the loss function as follow:

$$Loss = \frac{1}{T} \sum_{t \in T} (p_t - \hat{p}_t)^2. \quad (13)$$

where T is the training sets, p represents the true label, $\hat{p}$ is the model's predicted result and C represents the number of prediction task.

IV. EXPERIMENTS

A. Datasets

We obtain seven public datasets(BBBP [17], BACE [30], HIV [31], Clintox [32], Tox21 [33], SIDER [34], MUV [35]) and three regression datasets(FreeSolv [36], ESOL [37], Lipophilicity [38]) related to drug discovery from MoleculeNet [39]. These datasets encompass various molecular properties such as membrane permeability, toxicity, and biological activity [40]. Each dataset has a different number of molecules and subtasks. Specific information about the datasets is presented in Table I. Additionally, We evaluated the predictive ability of the AEGNN-M model using 14 public breast cancer cell line phenotype

TABLE II
PREDICTIVE PERFORMANCE RESULTS OF AEGNN-M ON SEVEN BENCHMARK DATASETS

Dataset	Split type	Metric	MoleculeNet (Graph)	chemprop	Attentive FP	HRGCN+	ATMOL+	FP-GNN	AEGNN-M
BBBP	Ramdom	ROC-AUC(%)	\	91.7	88.7	92.6	\	93.5	94.1
	Scaffold	ROC-AUC(%)	69.0(GC)	88.6	\	\	94.5	91.6	96.1
ClinTox	Ramdom	ROC-AUC(%)	83.2(Weave)	89.7	90.4	89.9	\	84.0	99.6
	Scaffold	ROC-AUC(%)	\	\	\	\	98.1	\	99.5
SIDER	Ramdom	ROC-AUC(%)	63.8(GC)	65.8	62.3	64.1	\	66.1	83.7
	Scaffold	ROC-AUC(%)	\	\	\	\	82.6	\	86.4
BACE	Ramdom	ROC-AUC(%)	\	89.8	87.6	89.1	\	88.1	86.7
	Scaffold	ROC-AUC(%)	80.6(Weave)	85.7	\	\	89.2	86.0	84.5
HIV	Ramdom	ROC-AUC(%)	\	82.7	82.2	82.4	\	82.5	79.4
	Scaffold	ROC-AUC(%)	76.3(GC)	79.4	\	\	83.2	82.4	80.0
MUV	Ramdom	PRC-AUC(%)	1.1(Weave)	5.3	3.8	8.2	\	9.0	1.0
Tox21	Ramdom	ROC-AUC(%)	82.9(GC)	85.4	85.2	84.8	\	81.5	84.9
	Scaffold	ROC-AUC(%)	\	\	\	\	82.5	\	83.1

GC: graph convolutional model;weave: weave model; bold font indicates the model that outperforms all other models.

TABLE III
PREDICTIVE PERFORMANCE RESULTS OF AEGNN-M ON CELL-BASED PHENOTYPIC SCREENING DATASETS

Cell lines	Split type	Compounds	Task metric	Attentive FP	GAT	GCN	MPNN	AEGNN-M
Bcap37	Random	275	ROC-AUC(%)	85.8	76.7	69.3	80.7	80.8
BT-20	Random	292	ROC-AUC(%)	73.5	72.1	74.0	78.4	91.5
BT-474	Random	811	ROC-AUC(%)	78.7	65.7	86.6	84.7	86.9
BT-549	Random	1182	ROC-AUC(%)	63.0	71.0	66.9	64.3	72.8
HBL-100	Random	316	ROC-AUC(%)	64.5	64.1	65.8	70.1	80.1
HS-578T	Random	469	ROC-AUC(%)	83.0	75.8	63.6	66.5	75.7
MCF-7	Random	29378	ROC-AUC(%)	84.5	80.0	83.3	84.3	81.2
MDA-MB-231	Random	11202	ROC-AUC(%)	87.0	77.0	85.9	85.0	90.7
MDA-MB-361	Random	367	ROC-AUC(%)	93.8	89.6	95.5	97.2	92.9
MDA-MB-435	Random	3030	ROC-AUC(%)	82.4	83.0	85.8	74.9	76.3
MDA-MB-453	Random	440	ROC-AUC(%)	87.2	81.2	86.6	71.5	87.8
MDA-MB-468	Random	1986	ROC-AUC(%)	87.5	87.5	88.7	85.8	94.7
SK-BR-3	Random	2026	ROC-AUC(%)	80.5	84.0	83.9	76.0	84.4
T-47D	Random	3135	ROC-AUC(%)	81.2	76.3	81.9	75.1	93.2
Average				80.9	77.4	79.8	78.19	84.9

Bold font indicates the model that outperforms all other models. The number of classification tasks for all datasets is 1.

screening datasets [41] in Table III, which included 13 breast cancer cell lines and one normal breast cell line.

B. Experimental Settings

we employed scaffold-based splitting and random splitting methods, creating an 8/1/1 train/validation/test subset split for each dataset. We utilized cross entropy loss for classification tasks, mean square error loss for regression tasks and set the learning rate to 1e-7. We employed the Adam optimizer. The performance of molecular property prediction was evaluated using ROC-AUC or PRC-AUC for classification tasks and RMSE for regression tasks. To prevent overfitting, we implemented early stopping and dropout strategies. The code, detailed settings and datasets are available on GitHub (<https://github.com/Sixseven-Five/AEGNN-M>).

C. Performance of the AEGNN-M Network Architecture on Seven Public Benchmark Datasets

Baselines: We compare AEGNN-M with baselines as follows:

- MoleculeNet includes GCN model and Weave model trained on molecular graph features [39].
- Chemprop utilizes graph convolutional networks to predict molecular properties [10].
- Attentive FP utilizes a combination of graph convolutional layers and attention mechanisms [42].
- HRGCN+ combining molecular graphs and molecular descriptors [43].
- ATMOL+ developed a novel molecular graph augmentation strategy to generate challenging positive samples [25].
- FP-GNN combined and simultaneously learned information from molecular graphs and fingerprints [24].

Experimental results are shown in Table II, where bold numbers represent the best-performing model for each specific task. We also have the following observations: (1) AEGNN-M performed the best in 8 tasks, including the random splitting tasks for BBBP, ClinTox, SIDER, and Tox21, as well as the scaffold splitting tasks for BBBP, ClinTox, SIDER, and Tox21. Chemprop performed the best in 2 tasks, including the random splitting tasks for BACE and HIV. Weave model performed the best in 1 task, which was the scaffold splitting

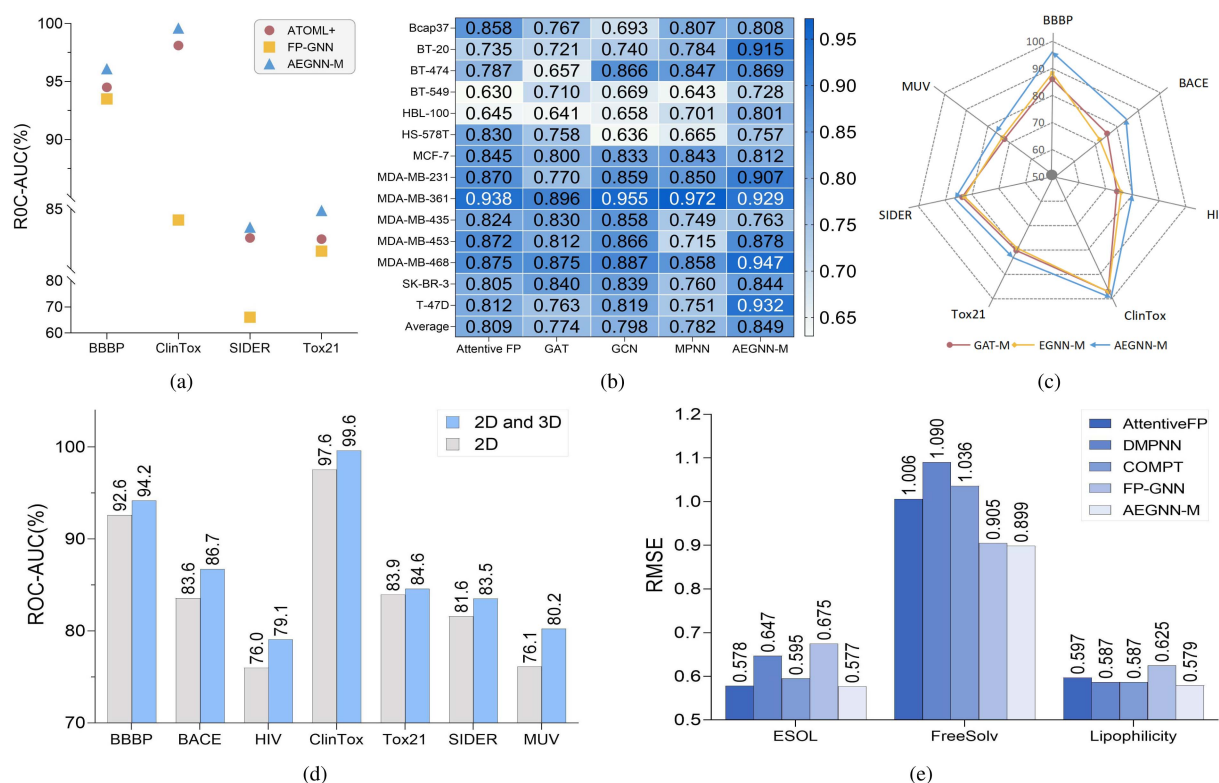


Fig. 3. Visualization of experimental results: (a) Figure compares the performance of the AEGNN model with ATOML+ and FP-GNN models. (b) Figure displays the performance evaluation results of the AEGNN-M model on the cell-based phenotype screening dataset. (c) Figure shows the ablation experiment results of the AEGNN-M model. (d) Figure presents the influence of 3D coordinate. (e) Figure presents the performance of the AEGNN-M model in three regression task.

task for MUV. ATOM+ model performed the best in 2 tasks, which were the scaffold splitting tasks for BACE and HIV. (2)The ultimate goal of building molecular property prediction models is to predict the properties of novel molecules with novel scaffolds. Fig. 3(a) shows the performance evaluation results of the AEGNN-M model, as well as the advanced ATOML+ and FP-GNN models, on four scaffold-based segmentation tasks: BBBP, Clintox, SIDER, and Tox21. It can be observed that the AEGNN-M model performs better than the ATOML+ and FP-GNN models. (3) These results collectively demonstrate the excellent and stable performance of the AEGNN-M model in predicting novel molecular properties. Furthermore, they indicate that combining 2D molecular graph representations with 3D molecular coordinate representations leads to better predictions of molecular characteristics.

D. Performance of the AEGNN-M Network Architecture on Three Regression Datasets

In addition to classification tasks, molecular property prediction tasks also include regression tasks. To evaluate the performance of the AEGNN-M model in the regression tasks. We compare AEGNN-M with state-of-the-art baseline models as follows:

- AttentiveFP adopts the graph attention network for molecular representation learning [41].

- DMPNN is a message passing model that takes into account edge features in a manner that involves interactive node-edge considerations [10].
- CoMPT model entails learning attributed graph representations through communicative message-passing transformers [13].
- FP-GNN integrated and acquired information from both molecular graphs and fingerprints [24].

In this study, we utilized three regression task datasets sourced from MoleculeNet. The experimental results are shown in Fig. 3(e), demonstrate that our model outperforms other model across all regression tasks. This substantiates the strong competitive edge of our model in regression tasks.

E. Performance of AEGNN-M on Cell-Based Phenotypic Screening Datasets

To evaluate whether the AEGNN-M model performs well on other datasets, we tested its performance on a phenotype screening dataset and compared the results with four graph-based deep learning models(Attentive FP, GCN, GAT, MPNN) reported by He et al. [41].

Experiment results are shown in Table III. Additionally, the results are visualized in the form of a heatmap in Fig. 3(b) Among the 14 cell lines. We also have the following observations:(1)AEGNN-M performed best on nine cell lines

TABLE IV
COMPARISON ROC-AUC(%) RESULTS OF THE PERFORMANCE BETWEEN THE AEGNN-M MODEL AND ITS PARTIAL MODULES GAT AND EGNN

model	split	BBBP	BACE	HIV	Clintox	Tox21	SIDER	MUV
GAT-M	Random	92.1	76.6	74.5	99.0	82.9	83.5	74.2
	Scaffold	86.0	75.6	74.2	97.0	80.2	83.7	72.3
EGNN-M	Random	92.1	80.1	74.4	96.3	81.9	83.6	76.1
	Scaffold	88.2	72.1	75.8	97.1	79.3	82.9	73.3
AEGNN-M	Random	94.1	86.7	79.4	99.6	84.9	83.7	80.2
	Scaffold	<i>96.0</i>	<i>84.5</i>	<i>80.0</i>	<i>99.5</i>	<i>83.1</i>	<i>86.4</i>	<i>76.3</i>

Bold font indicates the model that outperforms all other models in the random split task and italic font indicates the model that outperforms all other models in the scaffold split task.

(BT-20, BT-474, BT-549, HBL-100, MDA-MB-231, MDA-MB-453, MDA-MB-468, SK-BR-3, T-47D), the GCN model performed best on 4 cell lines (MDA-MB-231, MDA-MB-361, MDA-MB-435, T-47D), the Attentive FP model performed best on three cell lines (Bcap37, HS-578 T, MCF-7), and the GCN model performed best on two cell line (MDA-MB-361, MDA-MB-435). The results indicate that our AEGNN-M model performs exceptionally well on the phenotype screening dataset. (2) Among the five models, the AEGNN-M model exhibits the best overall average performance across the 14 cell lines, highlighting its substantial potential in phenotype-based drug discovery. Additionally, the model demonstrates stable performance in predicting molecular properties.

F. Ablation Experiment of AEGNN-M

To investigate the necessity of each module in AEGNN-M, we conducted several comparisons between AEGNN-M and its individual module on the independent test:

- AEGNN-M without EGNN only includes GAT module(GAT-M), 3D coordinate embedding feature no longer remains equivariant with changes in node features.
- AEGNN-M without GAT only includes EGNN module(EGNN-M), atom node features do not participate in the self-attention mechanism.

In this research study, we using seven widely used benchmark datasets from MoleculeNet. The experimental result indicates in Table IV, bold numbers represent the best-performing models for random segmentation tasks and italicized numbers represent the best-performing models for scaffold segmentation tasks. Fig. 3(c) displays a heatmap comparing the results of scaffold segmentation tasks. As we can see from Fig. 3(c), AEGNN-M outperformed GAT-M and EGNN-M in all other tasks. This observation highlights the exceptional predictive ability of AEGNN-M in determining the performance of novel scaffold molecules, surpassing the performance of its individual modules.

G. The Influence of 3D Coordinate

Different molecular representations possess different information and provide varying assistance in training models. The 2D molecular representation includes atomic features and edge features of the 2D molecular graph, while the 3D molecular representation includes atomic coordinates of the molecules in a 3D molecular graph. Therefore, the impact of the presence or

absence of 3D molecular representation on the performance of the AEGNN-M model architecture was investigated as follows.

- AEGNN-M model was tested using only the 2D molecular representation(2D).
- AEGNN-M model was tested using a combination of 2D and 3D molecular representations(2D and 3D).

The best performance advantages of the models based on 2D molecular representation and combined 2D and 3D molecular representation were statistically analyzed. The experimental results, as shown in Fig. 3(d), indicate that the mixed representation outperforms the single representation on each dataset. While 2D representations provide valuable information about the connectivity of atoms and bonds, they lack the ability to capture important structural features essential for accurate property prediction. Therefore, incorporating 3D molecular representation into the AEGNN-M model offers a more comprehensive and accurate representation of molecular structures, leading to enhanced performance on standard benchmark datasets.

V. THE INTERPRETATION OF AEGNN-M

A. Importance of Molecular Representations

In order to demonstrate the effectiveness of molecular representations in training the AEGNN-M model, we utilized UMAP [44] tool to visualize the molecular representation matrices before and after training. This visualization allows us to gain a clearer understanding of the model's performance. UMAP is a manifold learning algorithm that preserves the global structure of the data while effectively reducing its dimensionality. Fig. 4 presents the two-dimensional embeddings of molecules from the BBBP [17] and BACE [30] datasets. Initially, the molecular representations showcased a scattered and overlapping spatial distribution. However, after training, molecules belonging to the same class exhibited a tendency to cluster together, while also being noticeably separated from molecules of other classes. This observation suggests that our method accurately captures the physicochemical properties encoded in the molecular structures. Consequently, molecules with similar physicochemical properties obtained comparable latent representations, thereby highlighting the effectiveness of our approach.

B. Meaning of Attention Weights

In order to investigate the correlation between attention weights and molecular structures in the AEGNN-M model, the attention weights in the model were visualized. The BBBP

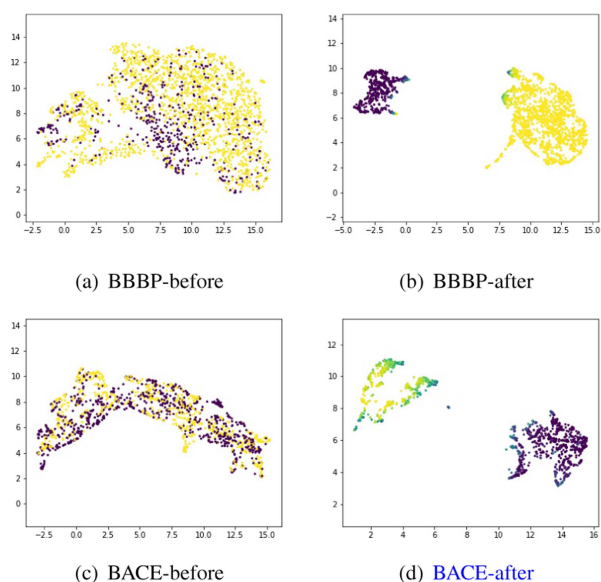


Fig. 4. Visualization of molecular representations in BBBP and BACE dataset. The left column showed the initial embeddings and the right column showed trained embeddings.

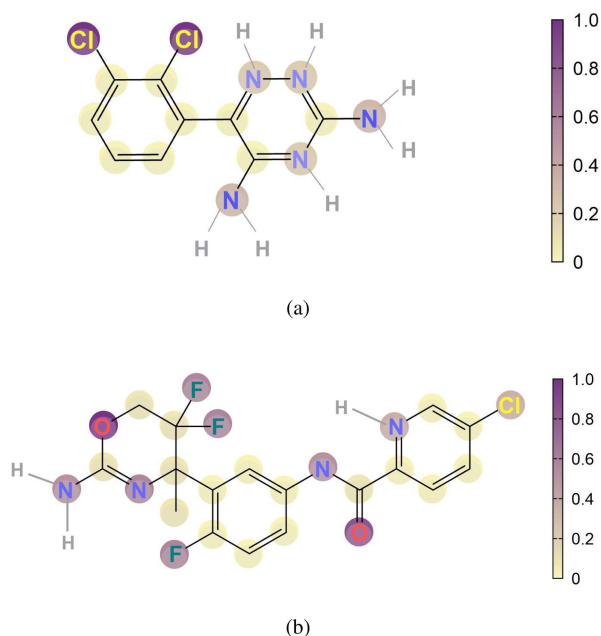


Fig. 5. Visualization of Model Attention Weights, (a) Molecule from the BBBP dataset, and (b) Molecule from the BACE dataset.

dataset is used to uncover the compound's blood-brain barrier permeability [45]. We selected a molecule from the BBBP dataset, Nc1nnc(c(N)n1)c2cccc(Cl)c2Cl, and visualized its attention weights in the model (Fig. 5a). It can be observed that each atom has different contribution to the overall molecule based on the attention weights. The Cl atom has the highest contribution throughout the molecule, while the N atom in the ring has a lower contribution compared to the N atom outside the ring, but still higher than the contribution of the C atoms. This indicates the importance of the Cl atom in the BBBP

dataset for a compound's ability to penetrate the blood-brain barrier. Similarly, BACE dataset is used to determine whether a compound exhibits high inhibitory activity against β -secretase 1. We also selected a molecule from the BACE dataset, Clc1ccc(nc1)C(=O)Nc1cc(C2=N=C(OCC2(F)F)N)C)c(F)cc1, and visualized its attention weights in the model (Fig. 5b). It can be observed that the O atom, F atom, and N atom have higher contributions to the overall model compared to the C atom. Additionally, the contribution of each C atom varies within the different rings, with the C atoms outside the benzene ring having higher.

VI. CONCLUSION

In this study, we propose a 3D graph-spatial co-representation model called AEGNN-M that enhances the integration of molecular representations and improves the accuracy of predicting molecular properties. AEGNN-M combines a graph self-attention network, based on 2D molecular graphs, with an equivariant neural network, which considers both 2D molecular graphs and 3D coordinates. We evaluate the performance of AEGNN-M against state-of-the-art deep learning methods on seven public datasets, three regression datasets and 14 breast cancer cell line phenotype screening datasets. The experimental results show that the AEGNN-M model performs significantly well. Moreover, we conduct a thorough analysis of the impact of different modules within AEGNN-M and the influence of spatial structural representations on its performance. The interpretability analysis also uncovers the importance of specific atoms in determining molecular properties. In the future, we will explore large molecular prediction models that have the ability to learn various molecular representations, whether they are 1D representation, 2D representation, or 3D representation.

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